

Практическая работа №7

ФИО: Комисарик Михаил Александрович

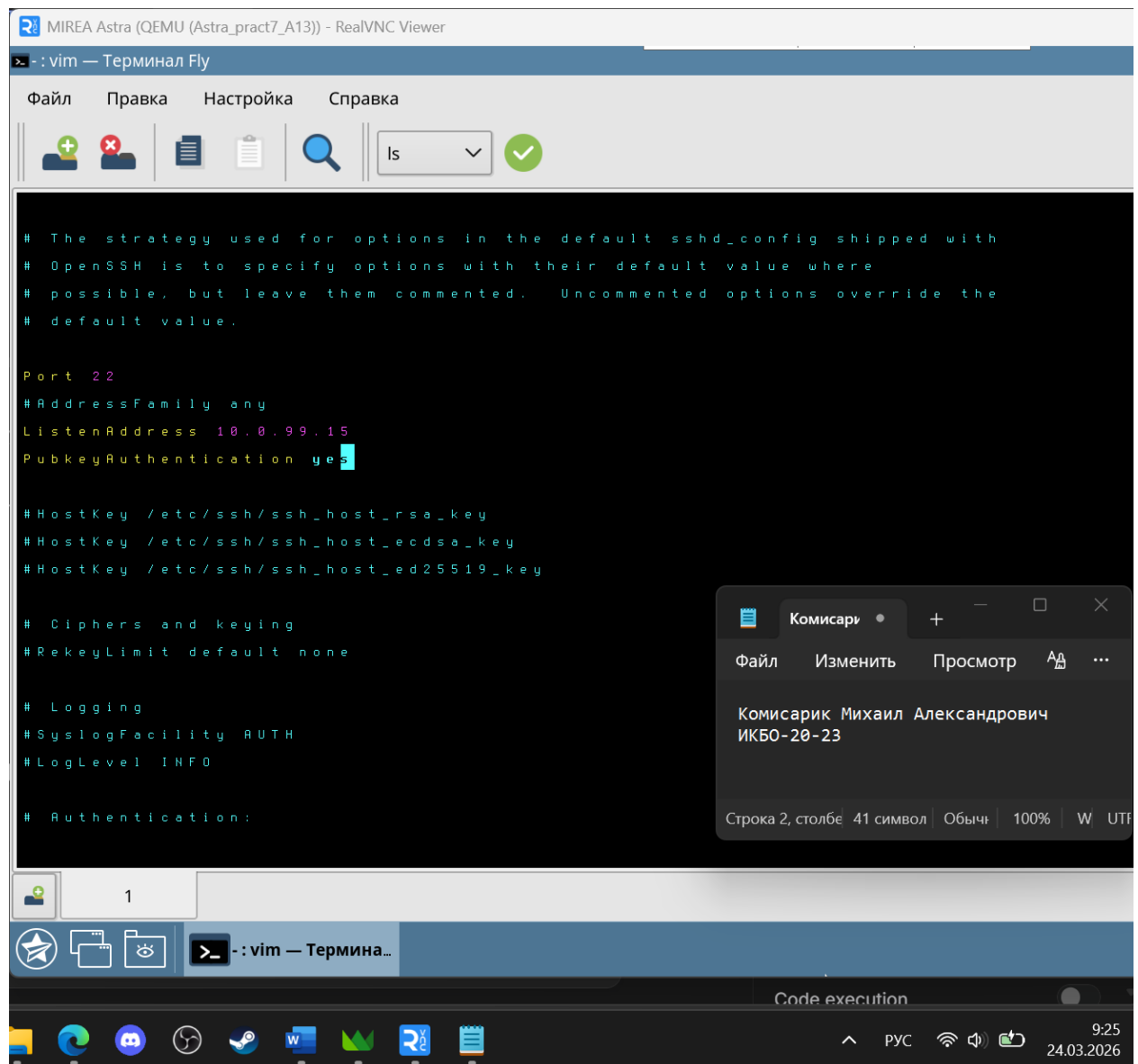
Группа: ИКБО-20-23

Данный отчет должен содержать скриншоты выполнения работы (замените скриншотом слово <..скриншот..> в соответствующем пункте).

На **ВСЕХ** скриншотах, которые вы делаете, должно быть видно ваше ФИО и группу (для этого откройте блокнот и запишите их там), текущую дату и время и номер ВМ.

Задание 1: Настройка Astra Linux

1. Конфигурация SSH



```
# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

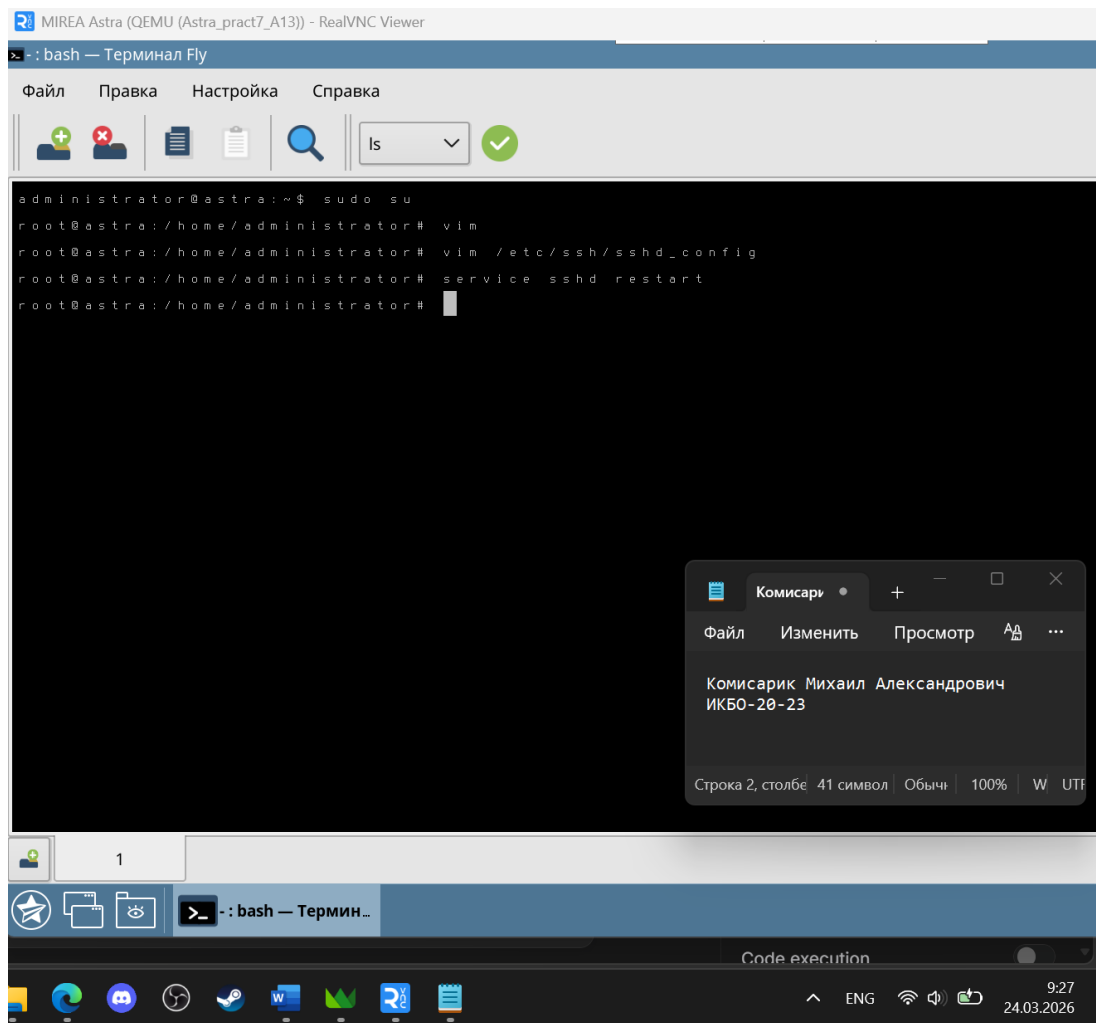
Port 22
#AddressFamily any
ListenAddress 10.0.99.15
PubkeyAuthentication yes

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

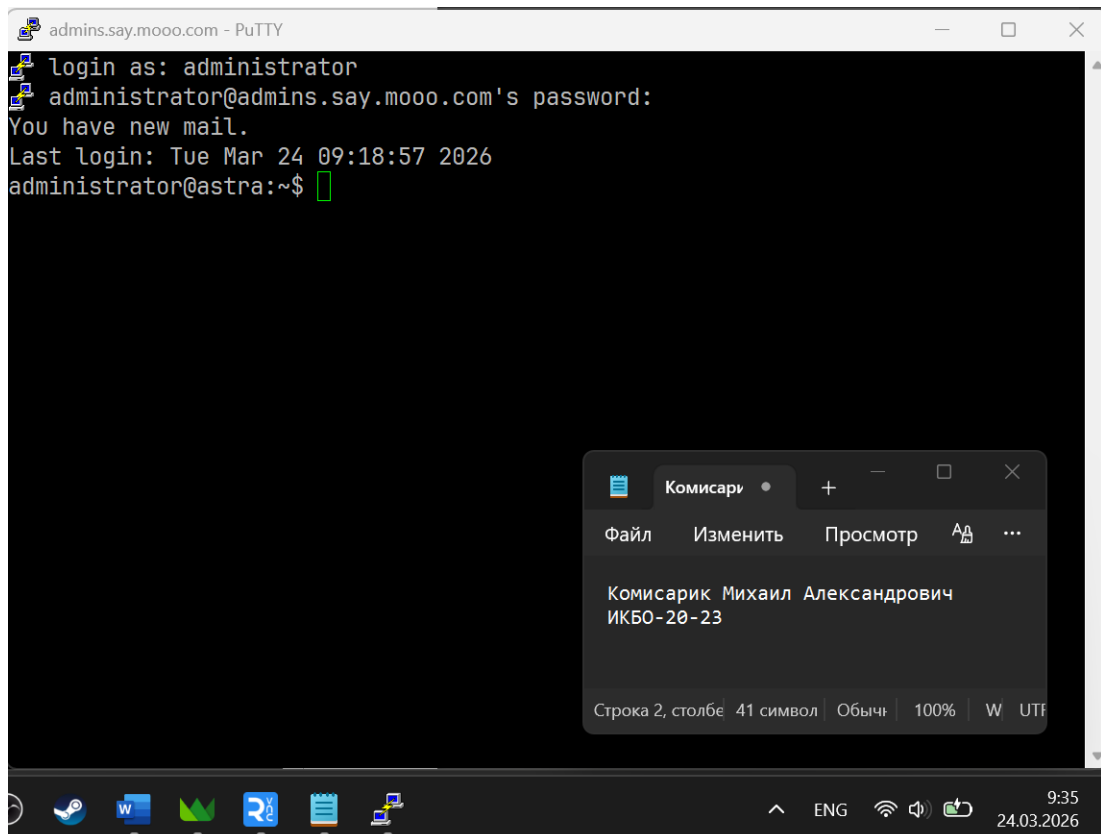
# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

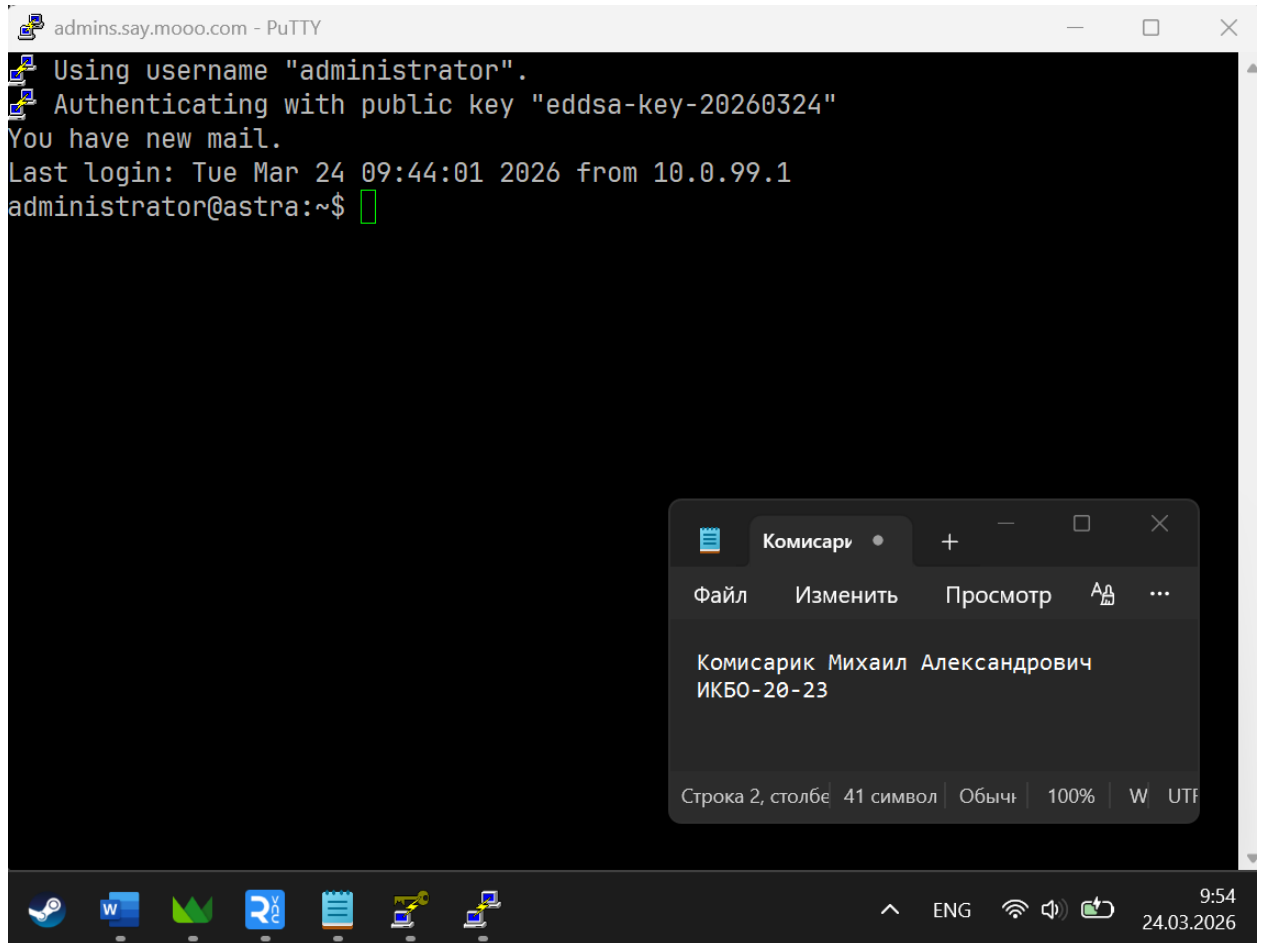
# Authentication:
```



2. Подключение через PuTTY



3. Подключение через PuTTY без пароля с помощью ключа генерации



4. Результаты базовой инвентаризации компьютера

processor : 0
vendor_id : GenuineIntel
cpu family : 6
model : 62
model name : Intel(R) Xeon(R) CPU E5-2697 v2 @ 2.70GHz
stepping : 4
microcode : 0x42d
cpu MHz : 2700.000
cache size : 16384 KB
physical id : 0
siblings : 1
core id : 0
cpu cores : 1

apicid : 0
initial apicid : 0
fpu : yes
fpu_exception : yes
cpuid level : 13
wp : yes
flags : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat
pse36 clflush mmx fxsr sse sse2 ss syscall nx pdpe1gb rdtscp lm constant_tsc
arch_perfmon rep_good nopl xtopology cpuid tsc_known_freq pni pclmulqdq vmx
ssse3 cx16 pdcm pcid sse4_1 sse4_2 x2apic popcnt tsc_deadline_timer aes xsave
avx f16c rdrand hypervisor lahf_lm cpuid_fault ssbd ibrs ibpb stibp tpr_shadow
vnmi flexpriority ept vpid fsgsbase tsc_adjust smep erms xsaveopt arat umip
flush_l1d arch_capabilities
vmx flags : vnmi preemption_timer posted_intr invvpid ept_x_only ept_1gb
flexpriority apicv tsc_offset vtptr mtf vapic ept vpid unrestricted_guest vapic_reg
vid shadow_vmcs
bugs : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds
swapgs mmio_unknown
bogomips : 5400.00
clflush size : 64
cache_alignment : 64
address sizes : 46 bits physical, 48 bits virtual
power management:

processor : 1
vendor_id : GenuineIntel
cpu family : 6
model : 62
model name : Intel(R) Xeon(R) CPU E5-2697 v2 @ 2.70GHz
stepping : 4

microcode : 0x42d
cpu MHz : 2700.000
cache size : 16384 KB
physical id : 1
siblings : 1
core id : 0
cpu cores : 1
apicid : 1
initial apicid : 1
fpu : yes
fpu_exception : yes
cpuid level : 13
wp : yes
flags : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat
pse36 clflush mmx fxsr sse sse2 ss syscall nx pdpe1gb rdtscp lm constant_tsc
arch_perfmon rep_good nopl xtopology cpuid tsc_known_freq pni pclmulqdq vmx
ssse3 cx16 pdcm pcid sse4_1 sse4_2 x2apic popcnt tsc_deadline_timer aes xsave
avx f16c rdrand hypervisor lahf_lm cpuid_fault ssbd ibrs ibpb stibp tpr_shadow
vnmi flexpriority ept vpid fsgsbase tsc_adjust smep erms xsaveopt arat umip
flush_l1d arch_capabilities
vmx flags : vnmi preemption_timer posted_intr invvpid ept_x_only ept_1gb
flexpriority apicv tsc_offset vtptr mtf vapic ept vpid unrestricted_guest vapic_reg
vid shadow_vmcs
bugs : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds
swapgs mmio_unknown
bogomips : 5400.00
clflush size : 64
cache_alignment : 64
address sizes : 46 bits physical, 48 bits virtual

power management:

dmidecode 3.0

Getting SMBIOS data from sysfs.

SMBIOS 2.8 present.

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state
UNKNOWN group default qlen 1000

link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00

inet 127.0.0.1/8 scope host lo

valid_lft forever preferred_lft forever

inet6 ::1/128 scope host

valid_lft forever preferred_lft forever

2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc
pfifo_fast state UP group default qlen 1000

link/ether 52:54:00:46:91:a4 brd ff:ff:ff:ff:ff:ff

inet 10.0.99.15/24 brd 10.0.99.255 scope global eth0

valid_lft forever preferred_lft forever

inet6 fe80::5054:ff:fe46:91a4/64 scope link

valid_lft forever preferred_lft forever

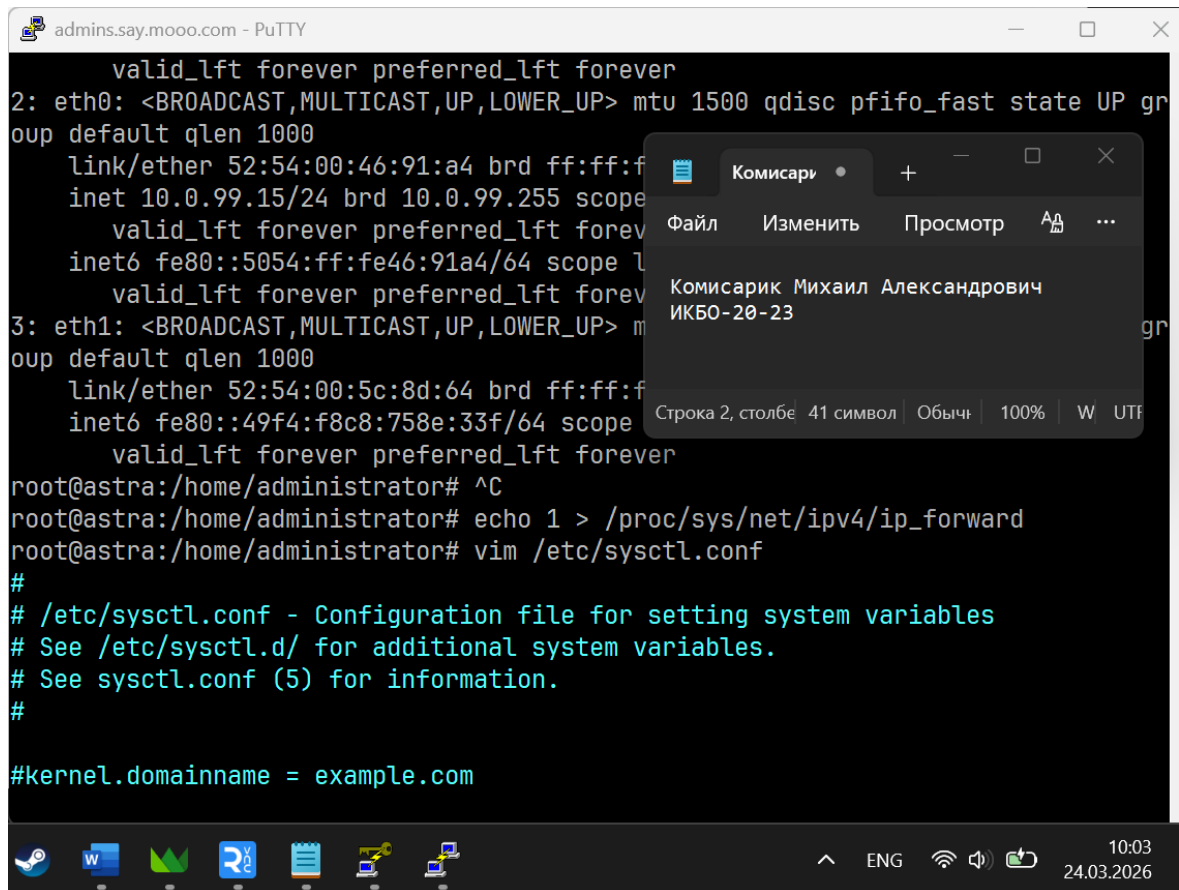
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc
pfifo_fast state UP group default qlen 1000

link/ether 52:54:00:5c:8d:64 brd ff:ff:ff:ff:ff:ff

inet6 fe80::49f4:f8c8:758e:33f/64 scope link noprefixroute

valid_lft forever preferred_lft forever

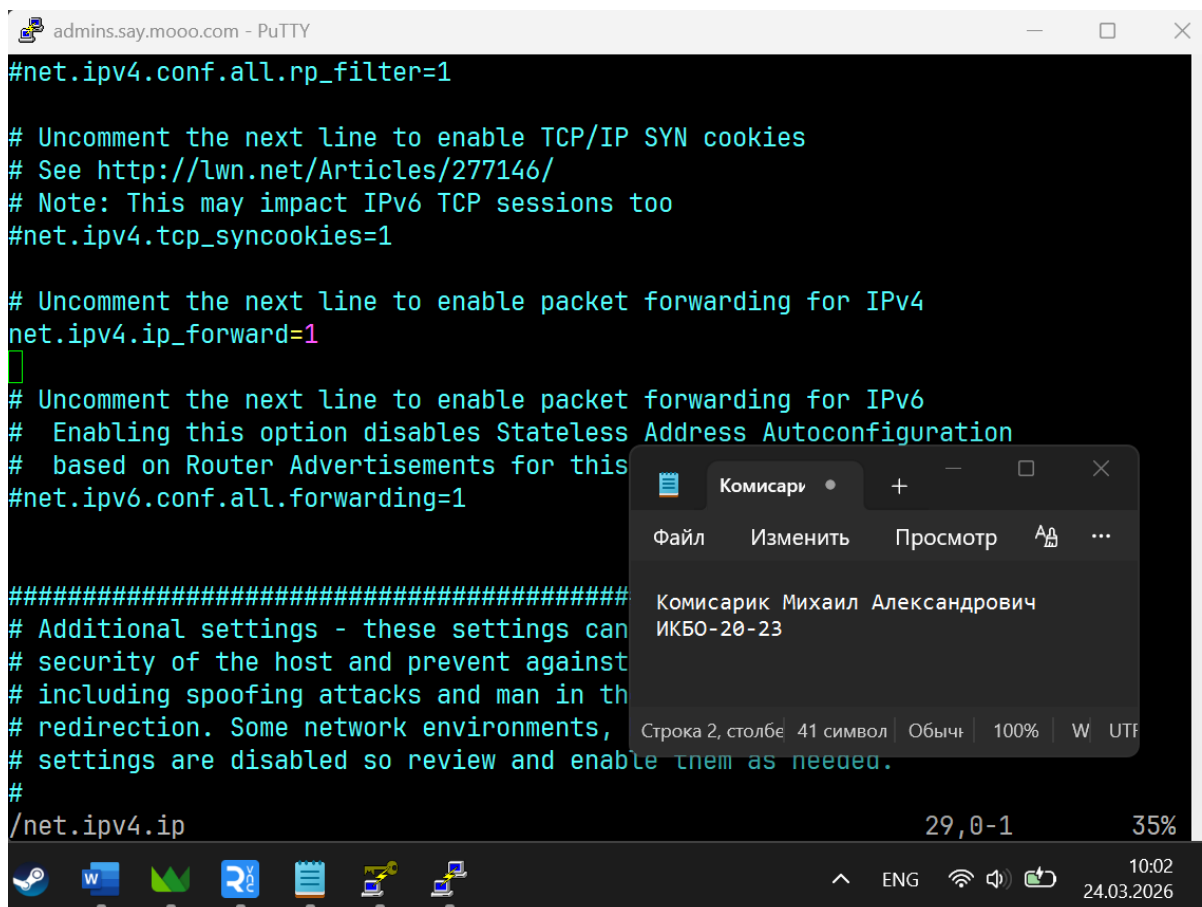
5. Включение перенаправления пакетов



The screenshot shows a PuTTY terminal window titled "admins.say.mo00.com - PuTTY". The terminal output includes network interface details for eth0 and eth1, followed by commands to set the IP forwarding flag and edit the sysctl configuration file. A vim editor window is overlaid on the terminal, showing the contents of /etc/sysctl.conf. The vim window has a title bar "Комисари" and a menu bar with "Файл", "Изменить", "Просмотр", and "A". The status bar at the bottom of the vim window shows "Строка 2, столбе 41 символ", "Обыч", "100%", "W", and "UTF".

```
valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP gr
oup default qlen 1000
    link/ether 52:54:00:46:91:a4 brd ff:ff:f
    inet 10.0.99.15/24 brd 10.0.99.255 scope
        valid_lft forever preferred_lft forev
    inet6 fe80::5054:ff:fe46:91a4/64 scope
        valid_lft forever preferred_lft forev
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> m
oup default qlen 1000
    link/ether 52:54:00:5c:8d:64 brd ff:ff:f
    inet6 fe80::49f4:f8c8:758e:33f/64 scope
        valid_lft forever preferred_lft forever
root@astra:/home/administrator# ^C
root@astra:/home/administrator# echo 1 > /proc/sys/net/ipv4/ip_forward
root@astra:/home/administrator# vim /etc/sysctl.conf
#
# /etc/sysctl.conf - Configuration file for setting system variables
# See /etc/sysctl.d/ for additional system variables.
# See sysctl.conf (5) for information.
#
#kernel.domainname = example.com
```

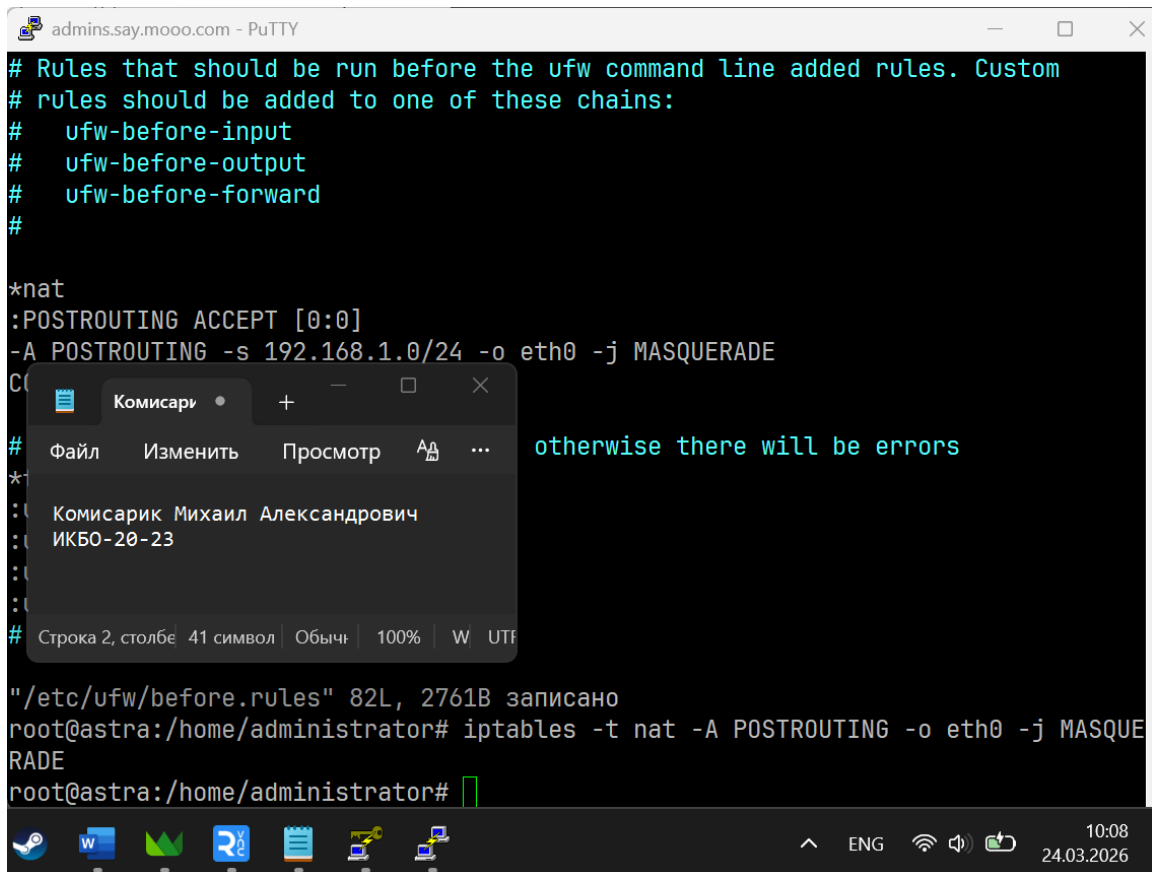
6. Отредактированный файл конфигурации /etc/sysctl.conf



The screenshot shows a PuTTY terminal window titled "admins.say.mo00.com - PuTTY". The terminal output shows the contents of the /etc/sysctl.conf file after editing. The vim editor window is overlaid on the terminal, showing the same title bar and menu bar as in the previous screenshot. The status bar at the bottom of the vim window shows "Строка 2, столбе 41 символ", "Обыч", "100%", "W", and "UTF".

```
#net.ipv4.conf.all.rp_filter=1
# Uncomment the next line to enable TCP/IP SYN cookies
# See http://lwn.net/Articles/277146/
# Note: This may impact IPv6 TCP sessions too
#net.ipv4.tcp_syncookies=1
# Uncomment the next line to enable packet forwarding for IPv4
net.ipv4.ip_forward=1
# Uncomment the next line to enable packet forwarding for IPv6
# Enabling this option disables Stateless Address Autoconfiguration
# based on Router Advertisements for this
#net.ipv6.conf.all.forwarding=1
#####
# Additional settings - these settings can
# security of the host and prevent against
# including spoofing attacks and man in th
# redirection. Some network environments,
# settings are disabled so review and enable them as needed.
#
/net.ipv4.ip
```

7. Включение маскардинга



```
admins.say.moood.com - PuTTY
# Rules that should be run before the ufw command line added rules. Custom
# rules should be added to one of these chains:
#   ufw-before-input
#   ufw-before-output
#   ufw-before-forward
#
*nat
:POSTROUTING ACCEPT [0:0]
-A POSTROUTING -s 192.168.1.0/24 -o eth0 -j MASQUERADE
COMMIT
# otherwise there will be errors
#
"/etc/ufw/before.rules" 82L, 2761B записано
root@astra:/home/administrator# iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
root@astra:/home/administrator#
```

Комисари

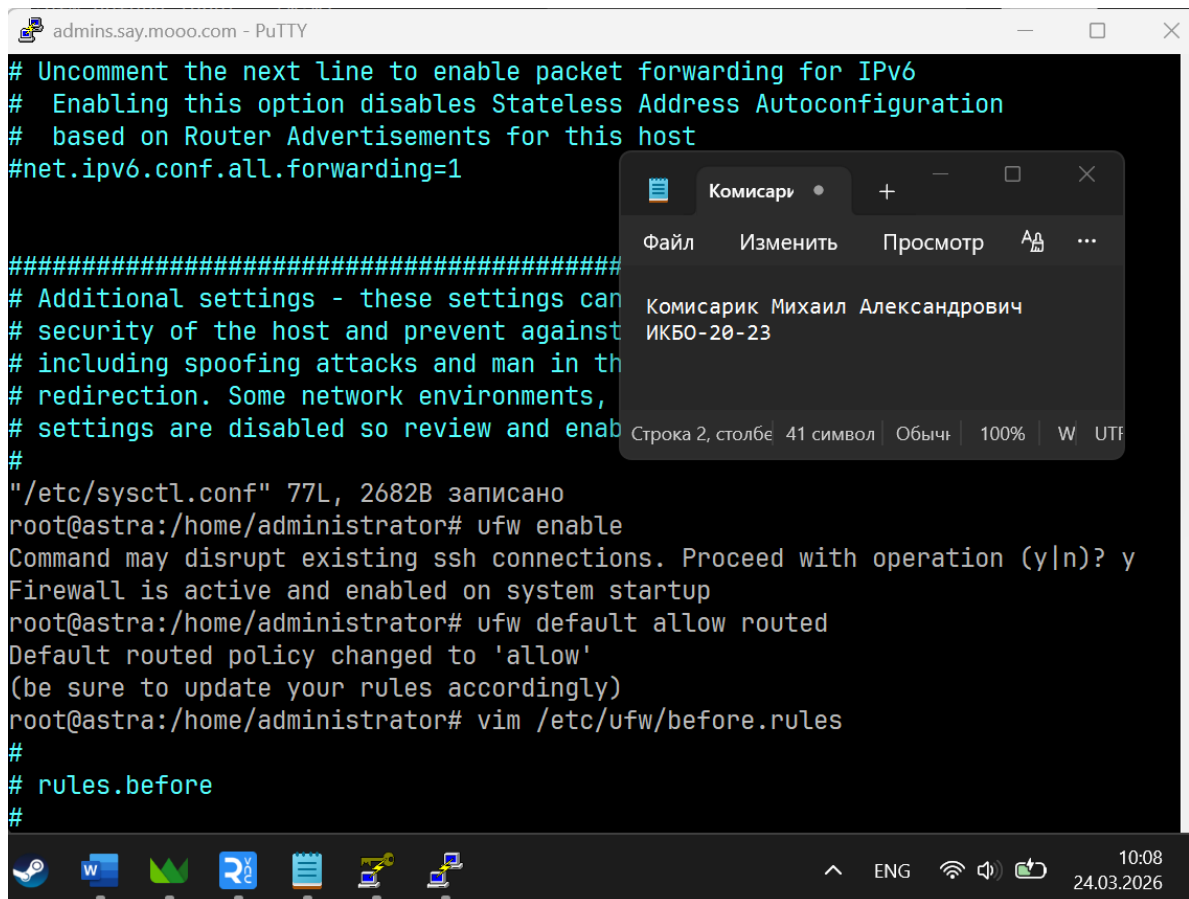
Файл Изменить Просмотр АА ...

Комисарик Михаил Александрович
ИКБ0-20-23

Строка 2, столбе 41 символ Обыч 100% W UTF

ENG 10:08 24.03.2026

8. Работа с ufw



```
admins.say.moood.com - PuTTY
# Uncomment the next line to enable packet forwarding for IPv6
# Enabling this option disables Stateless Address Autoconfiguration
# based on Router Advertisements for this host
#net.ipv6.conf.all.forwarding=1
#####
# Additional settings - these settings can
# security of the host and prevent against
# including spoofing attacks and man in the
# redirection. Some network environments,
# settings are disabled so review and enable
#
"/etc/sysctl.conf" 77L, 2682B записано
root@astra:/home/administrator# ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
root@astra:/home/administrator# ufw default allow routed
Default routed policy changed to 'allow'
(be sure to update your rules accordingly)
root@astra:/home/administrator# vim /etc/ufw/before.rules
#
# rules.before
#
```

Комисари

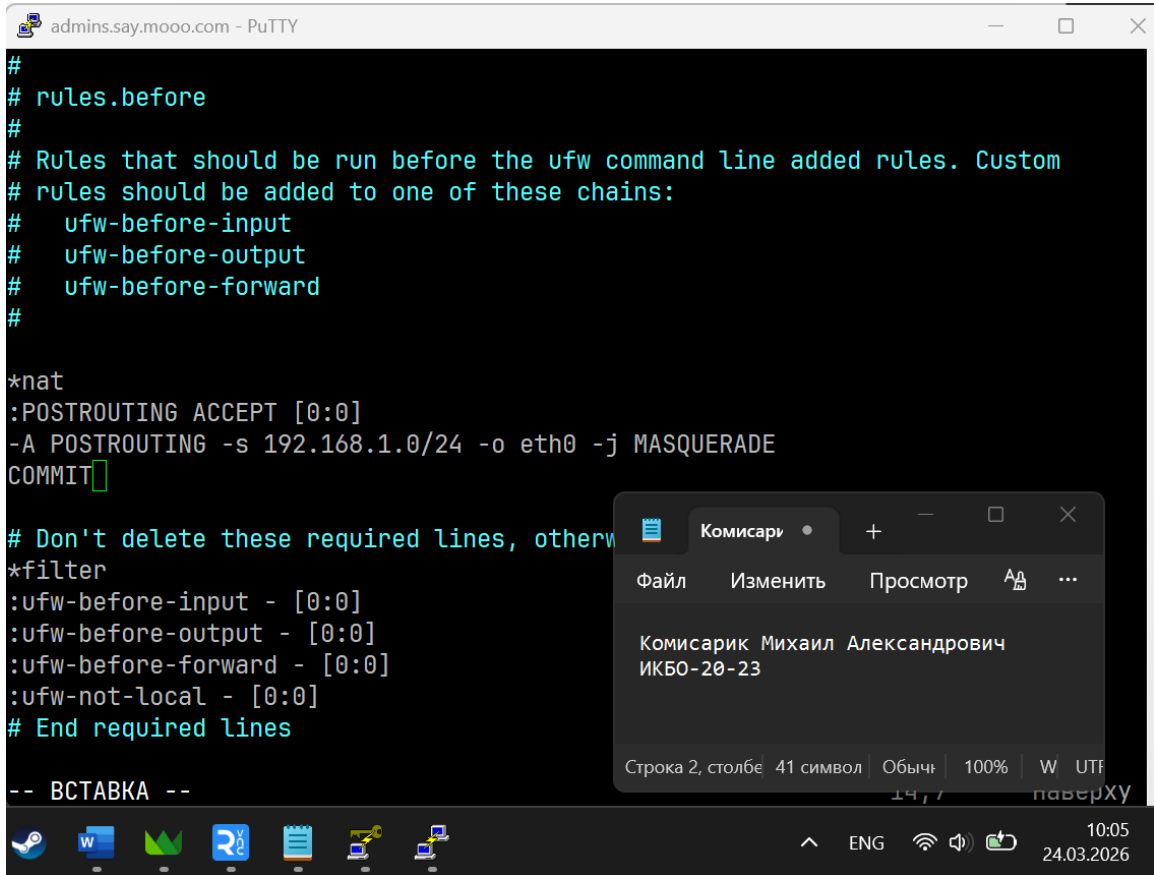
Файл Изменить Просмотр АА ...

Комисарик Михаил Александрович
ИКБ0-20-23

Строка 2, столбе 41 символ Обыч 100% W UTF

ENG 10:08 24.03.2026

9. Конфигурация /etc/ufw/before.rules

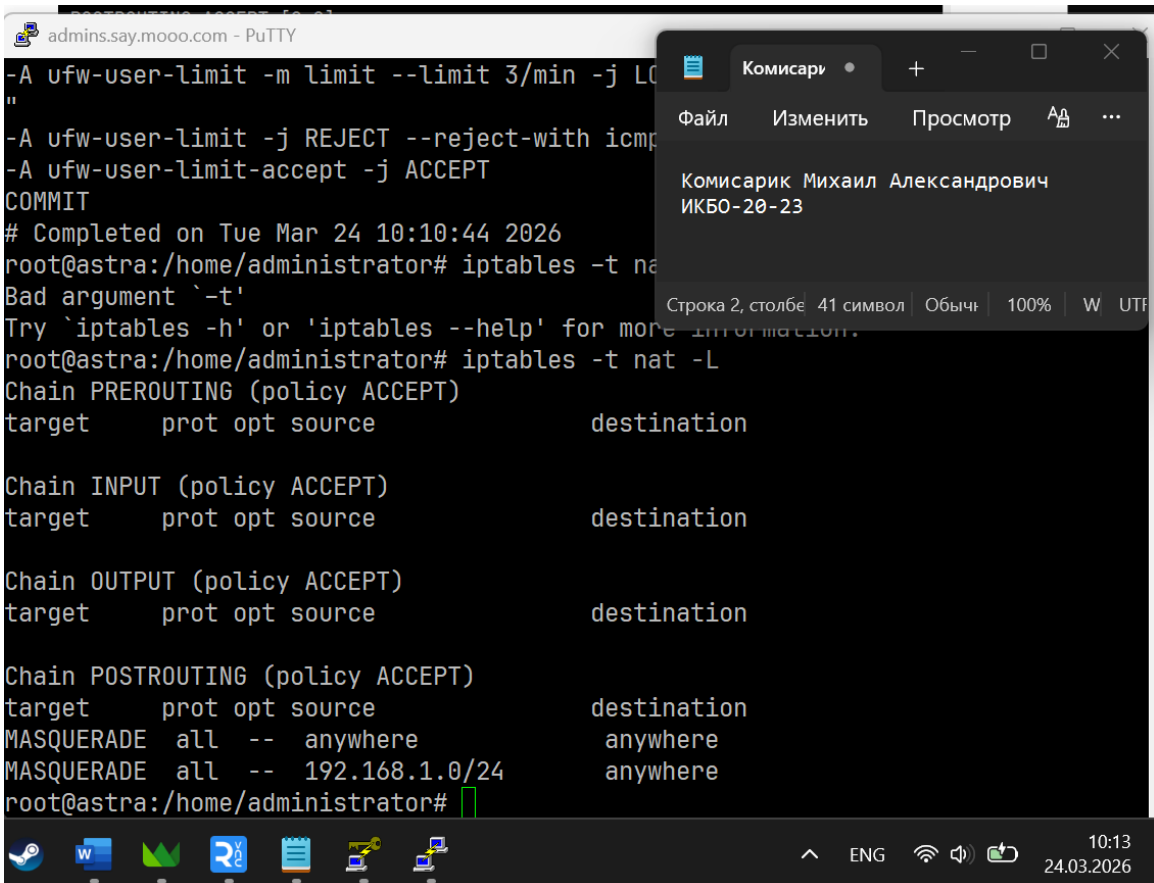


```
#
# rules.before
#
# Rules that should be run before the ufw command line added rules. Custom
# rules should be added to one of these chains:
#   ufw-before-input
#   ufw-before-output
#   ufw-before-forward
#
*nat
:POSTROUTING ACCEPT [0:0]
-A POSTROUTING -s 192.168.1.0/24 -o eth0 -j MASQUERADE
COMMIT

# Don't delete these required lines, otherwise
*filter
:ufw-before-input - [0:0]
:ufw-before-output - [0:0]
:ufw-before-forward - [0:0]
:ufw-not-local - [0:0]
# End required lines

-- ВСТАВКА --
```

10. Вывод iptables -t nat -L



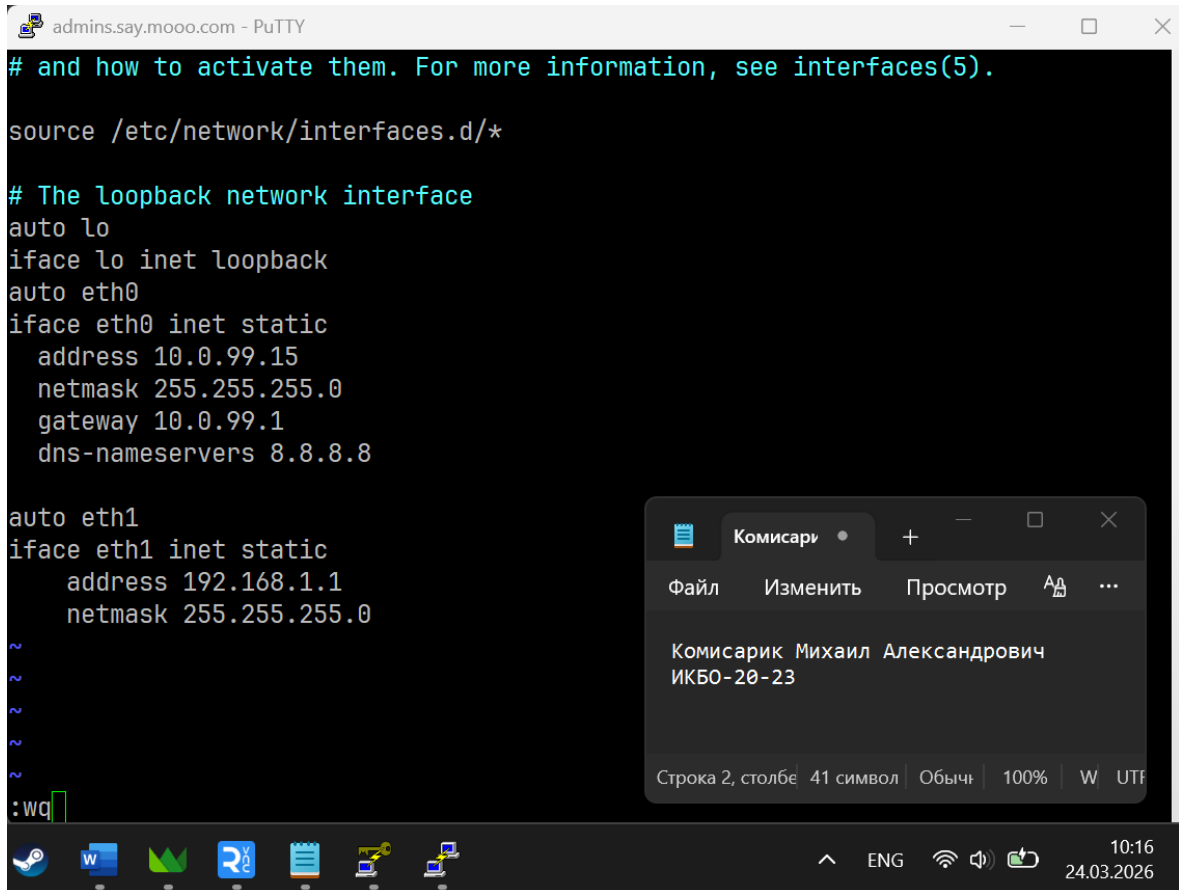
```
-A ufw-user-limit -m limit --limit 3/min -j LOG
"
-A ufw-user-limit -j REJECT --reject-with icmp
-A ufw-user-limit-accept -j ACCEPT
COMMIT
# Completed on Tue Mar 24 10:10:44 2026
root@astra:/home/administrator# iptables -t nat -L
Bad argument '-t'
Try 'iptables -h' or 'iptables --help' for more information.
root@astra:/home/administrator# iptables -t nat -L
Chain PREROUTING (policy ACCEPT)
target     prot opt source                destination

Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination

Chain POSTROUTING (policy ACCEPT)
target     prot opt source                destination
MASQUERADE all  --  anywhere              anywhere
MASQUERADE all  --  192.168.1.0/24        anywhere
root@astra:/home/administrator#
```

11. Настройка интерфейса eth1 (файл конфигурации)



```
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback
auto eth0
iface eth0 inet static
    address 10.0.99.15
    netmask 255.255.255.0
    gateway 10.0.99.1
    dns-nameservers 8.8.8.8

auto eth1
iface eth1 inet static
    address 192.168.1.1
    netmask 255.255.255.0
~
~
~
~
:WQ
```

Комисари

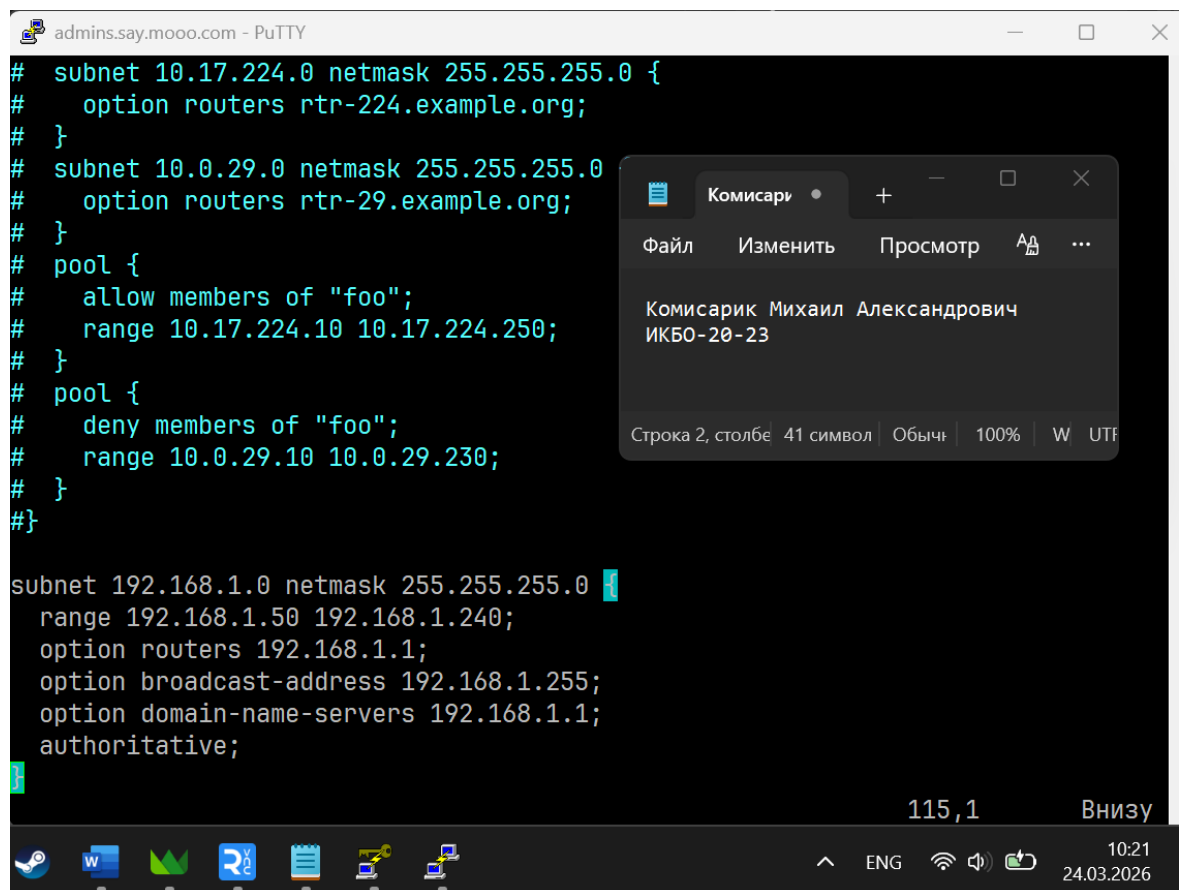
Файл Изменить Просмотр

Комисарик Михаил Александрович
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Строка 2, столбец 41 символ Обыч 100% W UTF

10:16
24.03.2026

12. Отредактированный файл конфигурации /etc/dhcp/dhcpd.conf



```
# subnet 10.17.224.0 netmask 255.255.255.0 {
#     option routers rtr-224.example.org;
# }
# subnet 10.0.29.0 netmask 255.255.255.0 {
#     option routers rtr-29.example.org;
# }
# pool {
#     allow members of "foo";
#     range 10.17.224.10 10.17.224.250;
# }
# pool {
#     deny members of "foo";
#     range 10.0.29.10 10.0.29.230;
# }
# }
#}

subnet 192.168.1.0 netmask 255.255.255.0 {
    range 192.168.1.50 192.168.1.240;
    option routers 192.168.1.1;
    option broadcast-address 192.168.1.255;
    option domain-name-servers 192.168.1.1;
    authoritative;
}
```

Комисари

Файл Изменить Просмотр

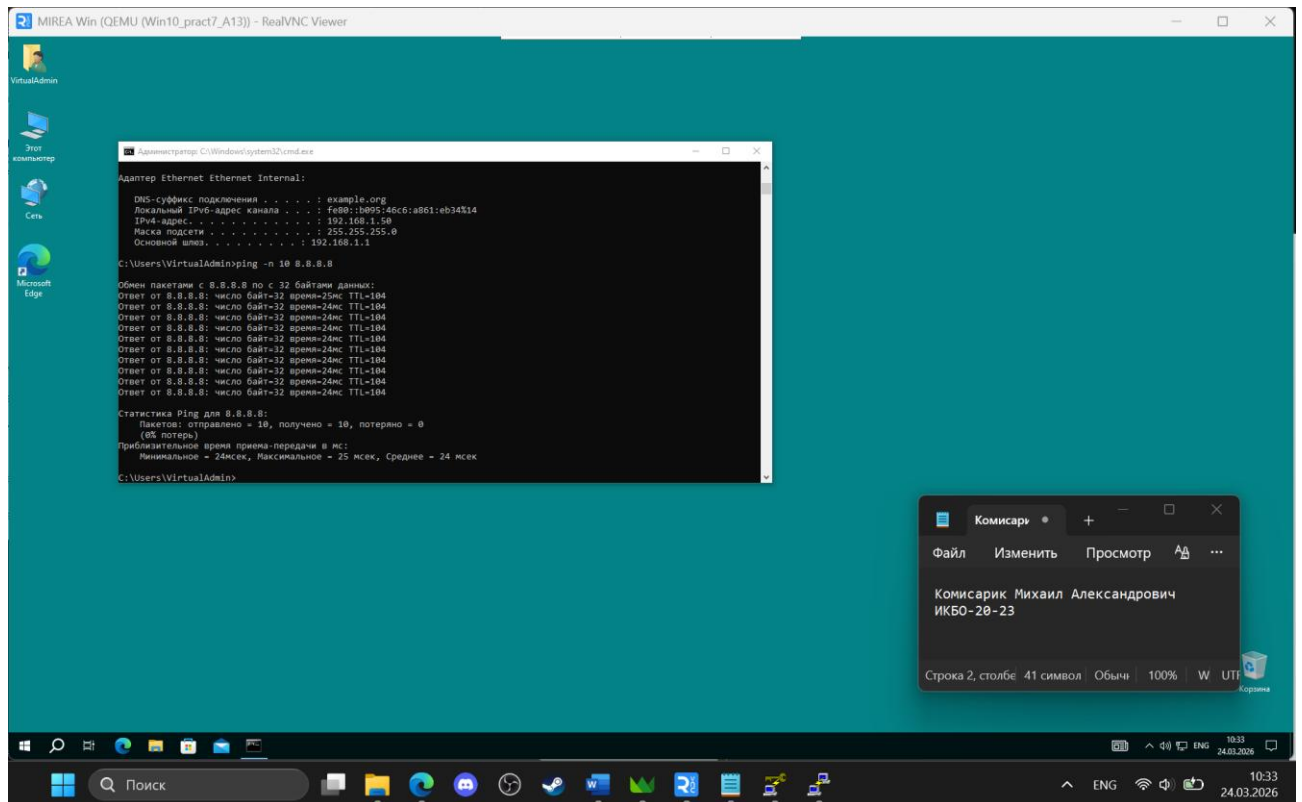
Комисарик Михаил Александрович
ИКБ0-20-23

Строка 2, столбец 41 символ Обыч 100% W UTF

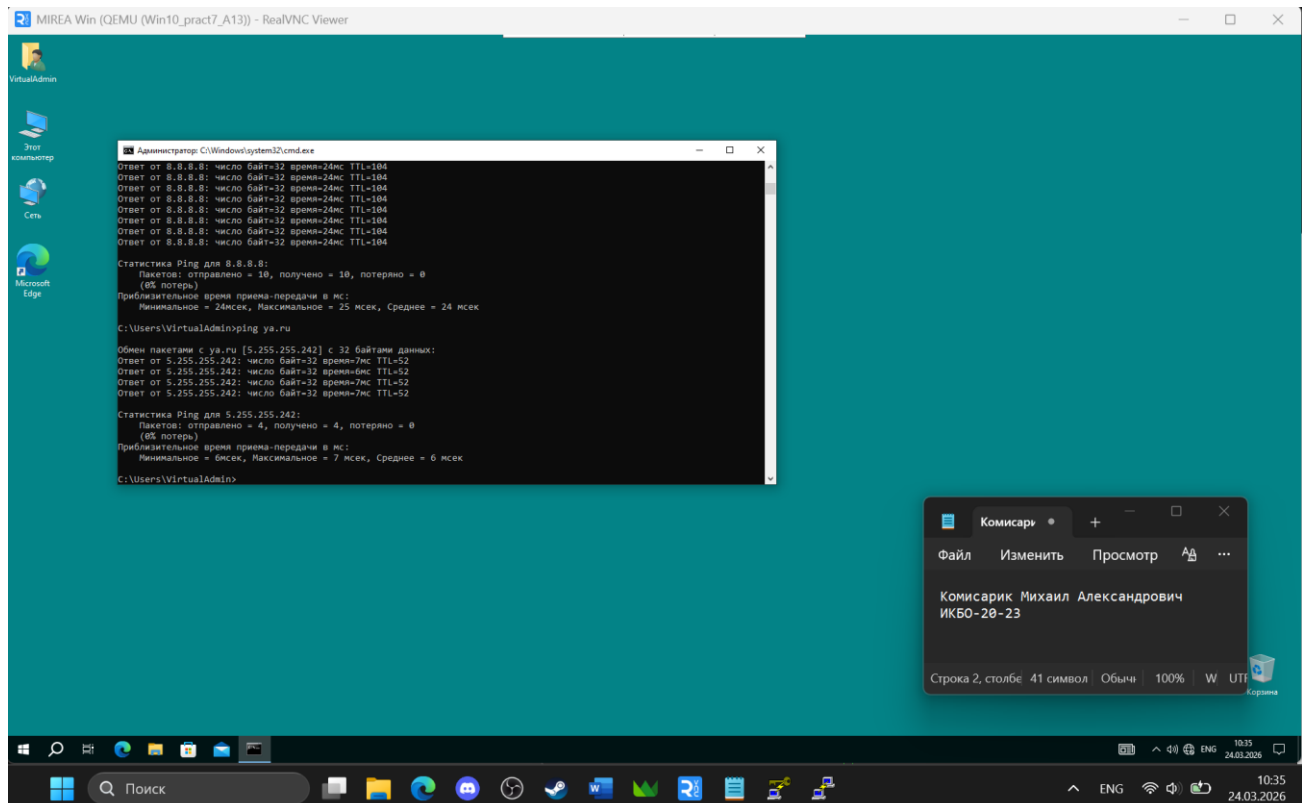
115,1 Внизу

10:21
24.03.2026

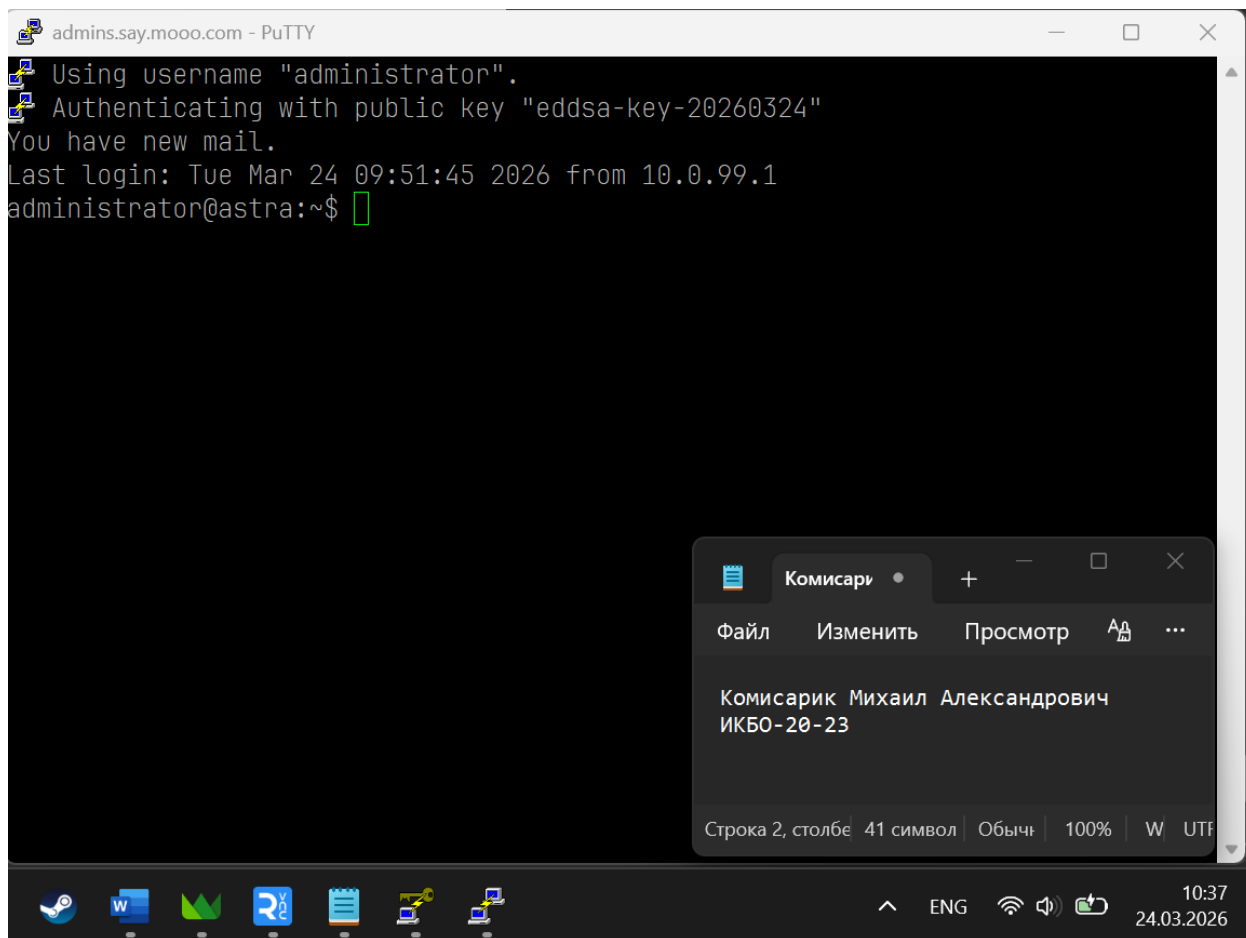
13. Проверка подключения к Интернету на VM с ОС Windows через ping 8.8.8.8



14. Проверка подключения к Интернету на VM с ОС Windows после настройки DNS через ping ya.ru



15. Подключение через PuTTY после настройки файервола



**Не забудьте выключить виртуальную машину после себя
(Пуск – Завершение работы).**